

STEP CUE · DESIGN DECISIONS REGISTER

StepCue design decisions register

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DOCUMENT CONTROL

Document ID	STC-DEC-001	Version	0.1
Status	Draft	Date	2026-10-01
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/stepcue		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-01	Amish Chadha	Register opened with the build plan; open items from STC-DDR-001 to STC-DDR-003 and the review note; budget treated as a value-engineering target

StepCue design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

StepCue is a research and educational prototype, not a medical device.

■ Open decisions

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Accept the design for construction changes P1 to P12 (two-layer laminate, routed traces with one crossover, sensor tabs on pads, motor through-hole, 9 mm slit tail, curved clip, tail connector in the pod, USB-C through the bottom wall, taped internal parts, screwed lid, button cap, heel notch)	(a) accept as modelled; (b) accept with changes	(a): each keeps what StepCue does, and the model passes 471 constructability checks	The whole build plan is drawn to these changes	STC-DDR-003, Table 1
2	How the insole and pod are charged	(a) insole and pod come out of the shoe together, as modelled; (b) an outside connector so the pod unplugs from the tail, adding a part and a wear point	(a) for the prototype; revisit after the first user session at TRL 4	Tail connector inside the pod (section 3.11, step 12)	STC-DDR-003, A1
3	Pod width and lid fixing	(a) 37 mm wide with four M2 screws; (b) 34 mm with a snap-fit lid and no screws	(a): screws are easier to get right first time; a snap lid can follow once the pod is printed	Pod base, lid, screws (sections 3.8 and 3.12)	STC-DDR-003, A2
4	Copper tape traces across the flexing arch and ball of the foot	(a) build as modelled and inspect after the first wear trials at TRL 4; (b) lay the traces in a zigzag where they cross the ball of the foot	(a): keep the first build simple; flex life is a TRL 4 test	Carrier film and traces (section 3.2)	STC-DDR-003, A3
5	First co-design partner	Movement disorders clinic; physiotherapy practice; Parkinson's patient group	None made	Not part of the TRL 3 build; needed before any user session	STC-DDR-001 and STC-DDR-002, O1
6	Appearance model: insole tilted 8 degrees toe-up in the hero render	Keep for the render only; drop	Keep for the render only; the flat fitted state stays the reference	Renders only	Review note, 2026-09-26, item 1
7	Smooth spline insole outline	Adopt in the model at the next revision; keep the polygon	Adopt at the next model revision	Insole outline template (Figure 4 of the build plan)	Review note, 2026-09-26, item 2
8	Appearance-only features (printed sensor rings, forefoot perforations, grip ribs, lid bezel recess, cadence mark)	Treat as appearance only; add to the model	Appearance only. The four M2 lid screws and their bosses are now in	Renders only	Review note, 2026-09-26, item 3

#	Decision needed	Options	Recommendation	Affects in the build	Source
			the model (STC-DDR-003)		
9	Rounded pod envelope (5 mm corner radius) with a parting-line groove	Adopt in the model; keep the square tray and flat lid	Adopt at the next model revision; it softens the edges near the wearer's heel	Pod base and lid shape	Review note, 2026-09-26, item 4
10	Clay shoe shell and foot as render context	Keep for renders; use in the concept media too	Keep for renders only	Renders only	Review note, 2026-09-26, item 5

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The pressure sensor's tail reaches about 12 mm past the disc and its two tabs are 2.54 mm apart	The tab pads on the carrier are placed for this; a longer tail moves them	STC-DDR-003, P2
2	How far the controller module's USB-C receptacle stands past its board end (1.4 mm assumed)	Sets whether a plug seats fully in the bottom-wall notch	STC-DDR-003, P8
3	The 8-way flat flex cable has same-side contacts and its 9 mm width matches the ZIF connector	The tail must go into the connector with its contacts toward the board	STC-DDR-003, P5 and P7
4	The cell is 502535 class (about 35 x 25 x 5 mm) and protected	The pod's inside is sized round it, and R15 needs a protected cell	STC-CAL-001, R10 and R15
5	The heel counter of the test shoe is about 3.5 mm thick and curved at about 32 mm radius inside	The clip's 1 mm grip and its curve are set for this counter	STC-CAL-001, R12
6	The polyimide film takes solder at about 300 °C without lifting its adhesive	The sensors, tail strips and motor leads are soldered to pads on it	STC-DDR-003, P1

■ Value engineering

Value-engineering target: USD 200 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 89.94 for one instrumented insole and one heel pod (USD 110.06 under the target). Main cost drivers and savings worth trying:

- The largest lines are the controller module (USD 15.99), the sensor laminate materials (USD 15.00), the five pressure sensors (USD 14.95), the cell (USD 7.00), and the interface board parts and hardware (USD 6.00 each).
- Making the design constructable added the tail connector (USD 2.50) and the spacer foam (USD 3.00) and repriced the hardware line (USD 5.00 to USD 6.00); the estimate rose from USD 83.44 to USD 89.94.
- Savings worth trying: buy laminate materials (polyimide film, copper tape) in quantities shared across several insoles, since most of the USD 15.00 is a minimum pack; a second insole for the other foot adds about USD 47 for the insole parts alone if it shares one pod design.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D8: one instrumented insole; nRF52840 module with built-in IMU; pressure plus IMU; electronics in a clip-on heel pod; haptic cue by default with audio through a phone or earbuds; rule-based detector first; commercial FSRs; requirement targets R1 to R15 adopted	Amish: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them."	STC-DDR-001
2026-09-25	O2: separate 0.3 s cue-start target for the phone or earbud audio route (0.1 s kept for haptic)	Amish: "i accept all your recommendations, go with them across all repos."	STC-DDR-002
2026-09-25	O3: 2.5 mm EVA insole base, 4.50 mm stack (the motor floor and laminate relief of that decision are since replaced by a through-hole, STC-DDR-003, open above)	Amish, same instruction	STC-DDR-002
2026-09-26	This repo chosen for the first batch of product renders	Amish	Review note, 2026-09-26
2026-09-30	Make the design physically buildable while drawing the build plan	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	STC-DDR-003 (draft, changes open above)
2026-10-01	Budget treated as a value-engineering target, not a limit	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens."	This register